

AN INTEGRATED BLOCKCHAIN AND CLOUD-BASED EXPLAINABLE AI FRAMEWORK FOR REDUCING BLACK BOX RISK IN DEEP LEARNING SYSTEMS

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ABSTRACT

Deep Learning (DL) models have demonstrated superior predictive capabilities across domains such as healthcare analytics, financial fraud detection, and intelligent transportation systems. Despite their high accuracy, their inherent black box behavior limits interpretability, transparency, and regulatory compliance, thereby reducing stakeholder trust in AI-driven decisions. Addressing this limitation is critical for deploying AI in high-risk and mission-critical environments.

This paper proposes an integrated framework that combines Cloud Computing, Explainable Artificial Intelligence (XAI), and Blockchain technology to mitigate black box risk in deep learning systems. The cloud layer enables scalable data storage, distributed model training, and real-time inference. An Explainable AI module is incorporated to generate interpretable insights using feature attribution and model explanation techniques. To ensure transparency and integrity, a blockchain layer is introduced to record model predictions, explanations, and version updates in an immutable and auditable manner.

The proposed framework is evaluated using metrics such as predictive accuracy, explanation fidelity, computational latency, and system security robustness. Experimental analysis demonstrates that the integration of explainability and decentralized audit logging significantly enhances transparency and trust without degrading model performance. The framework provides a scalable and secure foundation for developing trustworthy AI systems suitable for real-world deployment.

KEYWORDS

Explainable AI, Deep Learning, Blockchain, Cloud Computing, Trustworthy AI, Model Transparency, AI Auditing